VINUNIVERSITY

The W.A.T.E.Rs Live Map

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OUTLINE

1. General Information
2. System Description
3. Requirement Specification
4. Conclusions

1. GENERAL INFORMATION

- Purpose:
  - + Create a real-time mapping of water quality
  - + Measure water quality in distant area where people can't normally reach without wifi
  - + Store and display data in a server/ website
  - + Raise people awareness about local water quality
- Scope:
  - + Vinhomes Ocean Park
  - + Factories
  - + VinUniversity

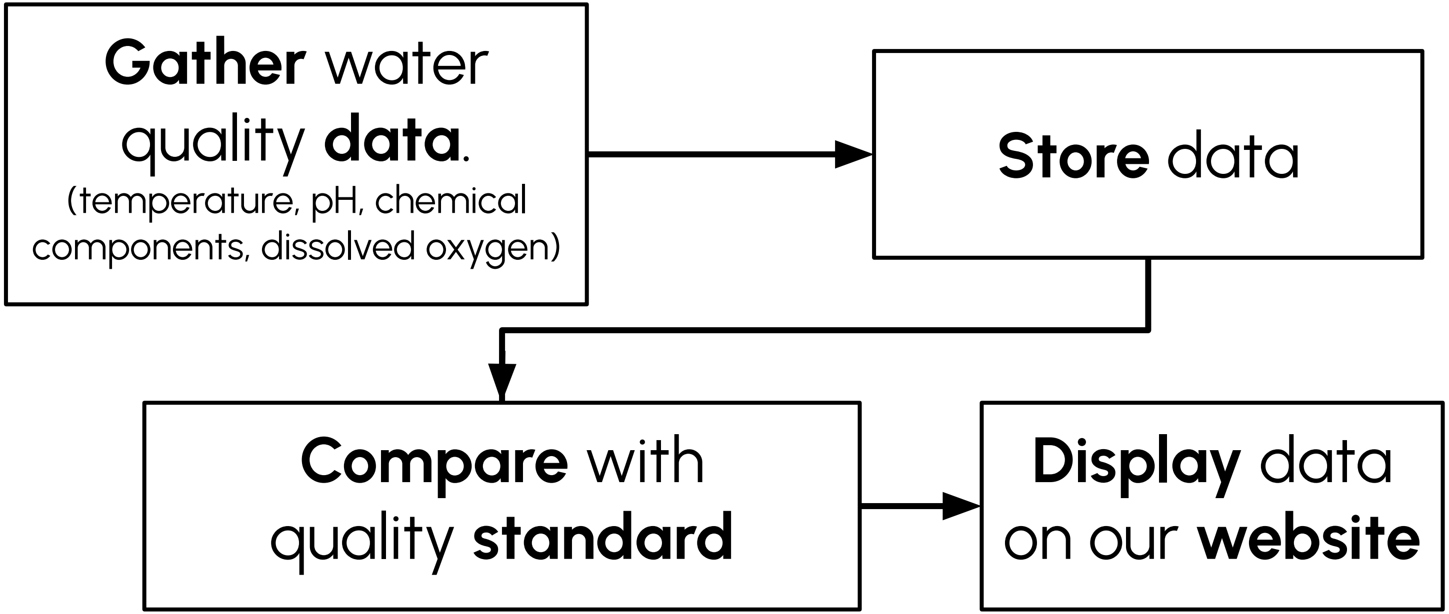
2. SYSTEM DESCRIPTION

- Structure of the system:
  - Water quality measure stations:
    - Mesh network via 2.4Ghz wave (ESP32)
  - Central Unit Control:
    - Router
    - Center Processor (Raspberry Pi)
  - Live Server:
    - Host on a server can be accessed via the Internet
  - Database
    - Cloud database with security rules (AWS, Firebase)

2. SYSTEM DESCRIPTION

- Use scenarios:
  - Particularly useful in faraway or inaccessible areas.
  - Capable of working for extended periods without human intervention.
  - Reduces the need for frequent maintenance and monitoring.
  - Daily updates water quality information to the database.
  - Ensures that the latest data is available for analysis.

2. SYSTEM DESCRIPTION



3. REQUIREMENT SPECIFICATION: Measuring

Functional Requirements	Non-functional Requirements
● Able to measure water quality at certain places. ● Provide data about color, freshness, temperature, pH, solid and dissolved components.	• Performance: Wifi must be 2.4GHz, able to transmit data in 5km range • Reliability: Sensor must catch data with +-0.01% accuracy • Durability: IP7. • Serviceability: Rechargeable and made of replaceable components • Conformance: Follow Vietnamese law. • Aesthetics: Light and portable. • Others (power consumptions, cost): 3-month operation for each charging, 1 million each block.

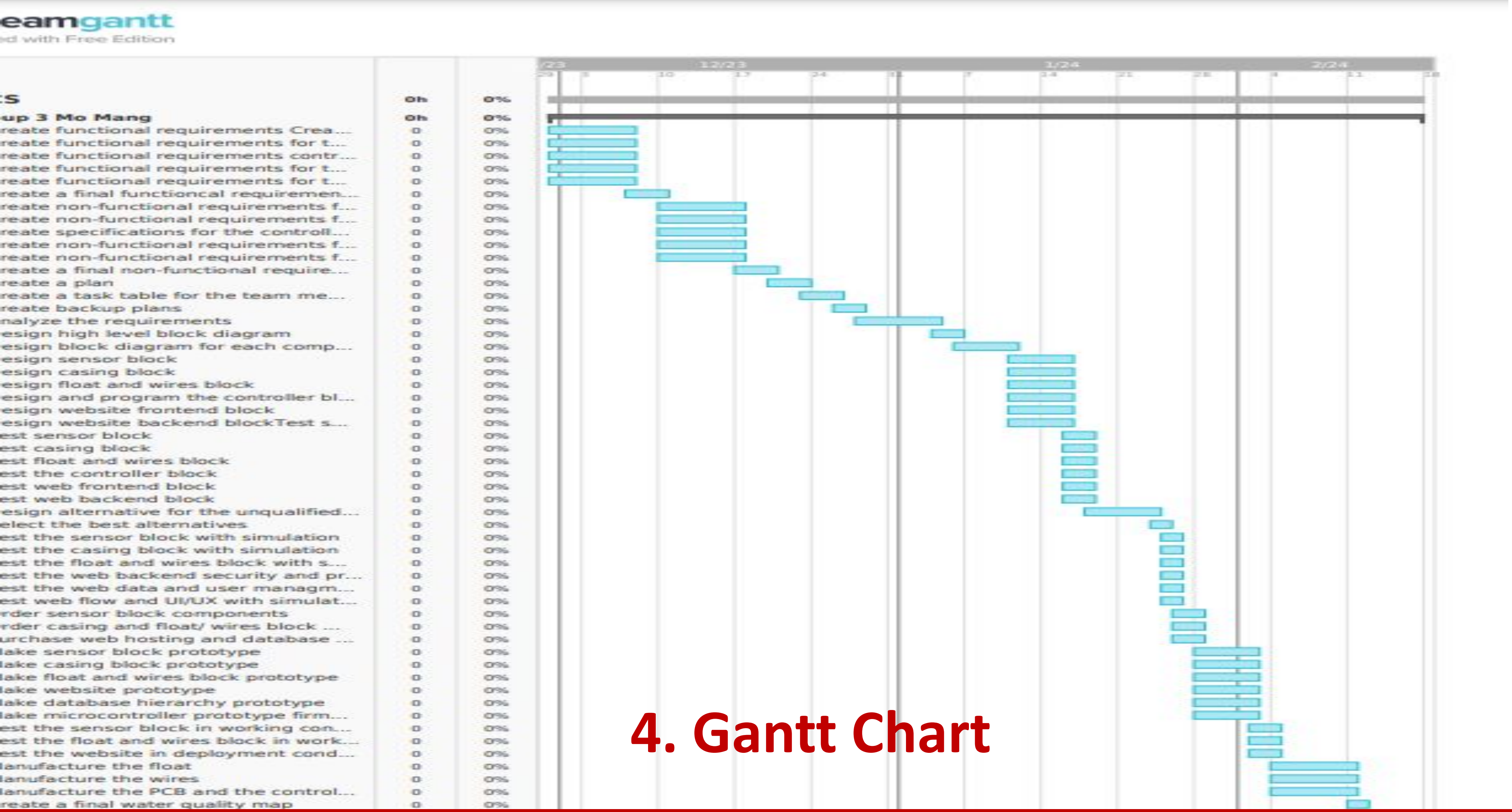
3. REQUIREMENT SPECIFICATION: Publishing

Functional Requirements	Non-functional Requirements
● Update regularly after each 2-hour period. ● Allow users to send reports and feedbacks on website. ● Able to provide user with basic solutions to prevent using low-quality water ● Map scaled of 1:1000 for Vietnam. ● Provide users with basic solutions to prevent using low-quality water. ● Provide interactive map functions	• Performance: Update instantly, track the time to refresh, • Reliability: Not taking ads promotion, non-profit • Durability: Restorable server • Serviceability: Having back-up data displayed when server got Ddos. • Conformance: Follow Vietnamese law. • Aesthetics: User-friendly and minimal interface. • Others (power consumptions, cost): 10$ hosting each year.

4. RACI

R = Responsible A = Accountable C = Consulted I = Informed

Task name	Dương Nghiệp Quý Electrical Engineering	Lê Hải Trung Mechanical Engineering	Hoàng Anh Thư Computer Science	Nguyễn Quốc Cường Computer Science	Trần Minh Khang Computer Science
Create functional requirements for the sensor block	R	A	I	I	I
Create functional requirements for the float and wires block	C	R	I	C	I
Create functional requirements controller block	R	R	I	I	I
Create functional requirements for the map UI/UX	C	R	I	I	I
Create functional requirements for the backend of the map	C	C	A	R	R
Create a final functioncal requirements for the whole system	I	I	R	R	A
Create non-functional requirements for the sensor block	C	I	A	C	R
Create non-functional requirements for the float and wires block	I	C	C	A	C
Create non-functional requirements controller block	I	I	A	R	R
Create non-functional requirements for the map UI/UX	C	R	I	A	I
Create non-functional requirements for the backend of the map	R	R	A	R	C
Create a final non-functional requirements for the whole system	R	R	C	A	I
Create a plan	C	C	A	R	R
Create a task table for the team members	I	I	R	R	A
Create backup plans	C	I	A	C	R
Analyze the requirements	I	C	C	A	C
Design high level block diagram	I	I	A	C	R
Design block diagram for each component	A	R	R	R	C
Design sensor block	R	A	I	I	I
Design casing block	C	R	I	I	I
Design float and wires block	R	R	I	C	I
Design and program the controller block	C	R	I	I	I
Design website frontend block	C	C	A	R	R
Design website backend block	I	I	R	R	A
Test sensor block	C	I	A	C	R
Test casing block	I	C	C	A	C
Test float and wires block	I	I	A	R	R
Test the controller block	C	R	I	A	I
Test web frontend block	R	R	A	R	C
Test web backend block	R	R	C	A	I
Design alternative for the unqualified blocks	C	C	A	R	R
Select the best alternatives	I	I	R	R	A
Test the sensor block with simulation	C	I	A	C	R



4. Gantt Chart

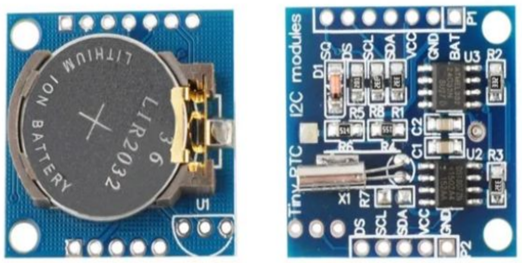
5. Progress

Priority	Status	Task Name	Description
High	Completed	Create functional requirements for the sensor block	Create specifications for the sensor block
High	Completed	Create functional requirements for the float and wires block	Create specifications for the float and wires block
High	Completed	Create functional requirements controller block	Create specifications for the controller block
High	Completed	Create functional requirements for the map UI/UX	Create specifications for the UI/UX
High	Completed	Create functional requirements for the backend of the map	Create specifications for the backend of the map
High	Completed	Create a final functoncal requirements for the whole system	Create a table of functional requirements for the whole system
High	Completed	Create non-functional requirements for the sensor block	Create specifications for the sensor block
High	Completed	Create non-functional requirements for the float and wires block	Create specifications for the float and wires block
High	Completed	Create non-functional requirements controller block	Create specifications for the controller block
High	Completed	Create non-functional requirements for the map UI/UX	Create specifications for the UI/UX
High	Completed	Create non-functional requirements for the backend of the map	Create specifications for the backend of the map
High	Completed	Create a final non-functional requirements for the whole system	Create a table of non-functional requirements for the whole system
High	Completed	Create a plan	Make a main plan for the product development
Medium	Completed	Create a task table for the team members	Split the task and assigned it to different members
Medium	Uncompleted	Create backup plans	Make backup plans in case the main plan failed
Medium	On-Going	Analyze the requirements	Analyze the requirements for each block
Medium	On-Going	Design high level block diagram	Create a high level workflow of the product
Medium	On-Going	Design block diagram for each component	Create the workflow for each block
Medium	On-Going	Design sensor block	Decide what components would be in the block and the combination of the components
Medium	Uncompleted	Design casing block	Decide on the shape of the case

Task Table

5. Progress

Real-time clock RTC DS1307



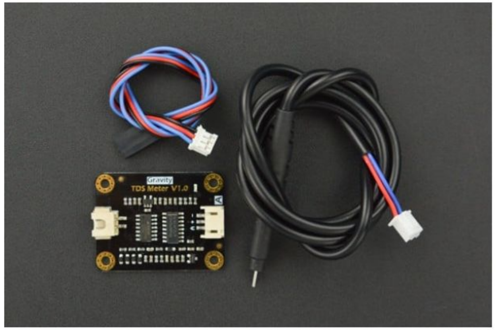
RTC to keep the real-time tick for sampling rate timing

ESP32C3 Seeed Studio

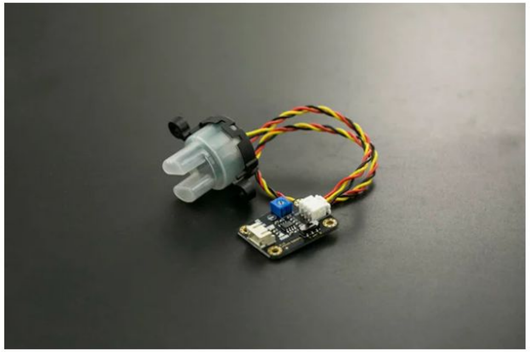
ESP32C3 Seeed Studio is a low power high connectivity microntroller with an ULP and 2.4Ghz Wifi antenna

5. Progress

Total Dissolved Solids Sensor DFRobot
Gravity: Analog TDS Sensor/Meter for Arduino



Water Turbidity Sensor DFRobot
Gravity: Analog TDS Sensor/Meter for Arduino



Wire Temperature Sensor DFRobot
Gravity: Analog TDS Sensor/Meter for Arduino

pH Sensor DFRobot
Gravity: Analog TDS Sensor/Meter for Arduino

6. CONCLUSIONS

- A high-tech system aimed to automatize water quality control process, reducing anthropic errors, providing further optimization, and increasing administrative power.
- Provide a qualified database for multi-purpose usage.
- Can cover a wide range of interested region, able to take measurement from distant places.